NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 23/03/2026
Time: 3 Hrs | GTM – 02 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
A	B	A	A	A	A	D	C	A	B
11	12	13	14	15	16	17	18	19	20
D	C	A	C	B	B	D	A	A	A
21	22	23	24	25					
15	24	69	16	100					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
D	C	C	D	B	D	C	C	D	D
36	37	38	39	40	41	42	43	44	45
D	C	A	D	A	C	C	B	A	C
46	47	48	49	50					
9	21	75	4	6					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	C	D	C	C	B	B	C	A	A
61	62	63	64	65	66	67	68	69	70
B	D	C	B	D	C	A	B	B	A
71	72	73	74	75					
9	5	320	30	27					

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HINTS
PHYSICS

1. $\frac{\Delta R}{R}\times 100=\frac{\Delta A}{A}\times 100+3\frac{\Delta C}{C}\times 100+2\frac{\Delta B}{B}\times 100$

$\frac{\Delta R}{R}\%=0.25\%+0.2\%+0.75\%=1.2\%$

2. For maximum intensity on the screen

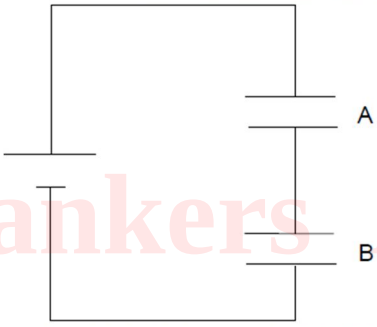
$d\sin\theta=n\lambda$

$\Rightarrow \sin\theta=\frac{n\lambda}{d}=\frac{n(2000)}{7000}=\frac{n}{3.5}$

Since maximum values of $\sin\theta$ is 1

So $n=0,1,2,3$ only. Thus, only seven Maximas can be obtained on both sides of the screen.

3.



when separation becomes double capacitance of A and also equivalent capacitance decreases there line charge on both A and B decreases.

4. (D)

5. On increasing temperature of ball by $100^{\circ}C$ (from $20^{\circ}C$ to $120^{\circ}C$), the thermal expansion in its volume can be given as

$\Delta V=V\gamma_s\Delta T=V(3\alpha_s)\Delta T.....(1)$

Since it is given that no change is volume is allowed, so the increment in volume due to thermal expansion is cancelled due to elastic compression by external pressure.

The bulk's modulus of a material is defined as

$B=\frac{\Delta P}{\Delta V/V}$

The excess pressure inside the ball at $120^{\circ}C$ to keep its volume constant during the heating process is

$\Delta P=B\left(\frac{\Delta V}{V}\right)=B(3\alpha_s\Delta T)$

$\Rightarrow \Delta P=(1.6\times 10^{11})[3(1.1\times 10^{-5})(100)]$

$\Rightarrow \Delta P=5.28\times 10^{8}\ Nm^{-2}=5.28\times 10^{8}\ Pa$

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6. $L \cdot \frac{di_L}{dt} = t^2$

$2 \cdot \int_0^{i_L} di_L = \int_0^2 t^2 dt$

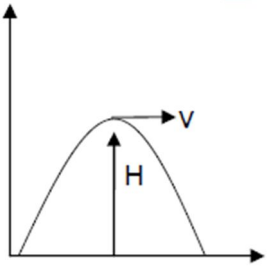
$2i_L = \frac{8}{3} \Rightarrow i_L = \frac{4}{3} A$

$i_c = \frac{d}{dt}\left(2t^2\right) = 4t$

$i_c = 8\ A$

$i_R = 2\ A$

7.



$mvH = L$

$QE_0H = \frac{1}{2}mv^2$

$H = \left(\frac{L^2}{2mE}\right)^{\frac{1}{3}}$

8. The coefficient of viscosity is the ratio of tangential stress on top surface of film (exerted by block) to that of velocity gradient (vertically downwards) of film. Since mass m moves with constant velocity, the string exerts a force equal to mg on plate towards right. Hence oil shall exert tangential force mg on plate towards left.

$\therefore\ \eta = \frac{F/A}{(v-0)/\Delta x} = \frac{125 \times 1000/10 \times 20}{(5-0)/.02} = 2.5$ dyne–s/cm²

9. . For Balmar series

$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n_i^2}\right)$

When $n_i \to \infty, \frac{1}{\lambda_{min}} = \frac{R}{4}$

$\Rightarrow \lambda_{min} = \frac{4}{R}$

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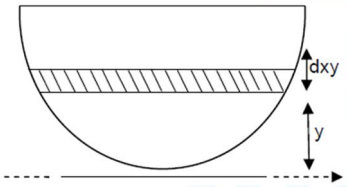
When $n_i = 3, \frac{1}{\lambda_{max}} = R\left(\frac{5}{36}\right)$

$$\lambda_{max} = \frac{36}{5R}$$

$$\frac{\lambda_{min}}{\lambda_{max}} = \frac{(4/R)}{(36/5R)} = \frac{20}{36}$$

$$\frac{\lambda_{min}}{\lambda_{max}} = \frac{5}{9}$$

10.



$$\phi = \int \frac{\mu_0 i}{2\pi y} \cdot 2\sqrt{y}\, dy$$

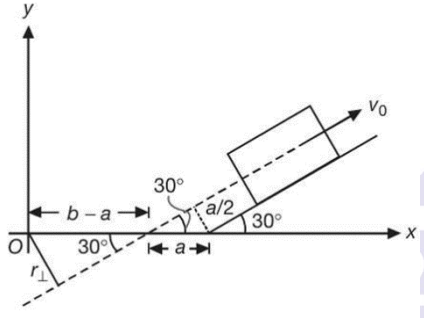
$$\phi = \frac{\mu_0 i}{\pi} \int y^{-\frac{1}{2}} dy$$

$$= \frac{2\mu_0 i}{\pi}\left[\sqrt{y}\right]_0^{\frac{\ell^2}{4}}$$

$$\phi = \frac{2\mu_0 i}{\pi} \cdot \frac{\ell}{2}$$

$$M = \frac{\mu_0}{\pi} \cdot \ell$$

11. $r_\perp = (b-a)\sin 30^\circ = \frac{b-a}{2}$



$$\Rightarrow L = mv_0 r_\perp = \frac{mv_0(b-a)}{2}$$

12. The speed is due to radial motion as well as due to angular motion so we use

$$v = \sqrt{\left(\frac{dR}{dt}\right)^2 + (R\omega)^2}$$

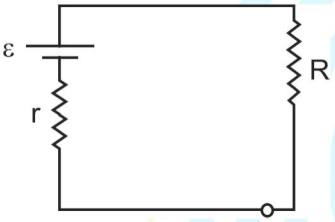
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$= \sqrt{\beta^2 + (R_0 + \beta t)^2}$

13. (i) λ_1 is of radio waves
(ii) λ_2 is of UV
(iii) λ_3 is of X-ray
(iv) λ_4 is of infrared

14. $I_{Z,max} = 25mA$, $I_L = 15mA$
$V_{i,max} = 30 + IR = 38$ V (I is the current passing through series resistance)
$V_{i,max} = 30 + IR = 38$ V
$I_{Z,min} = 0$, $I_L = 15mA$
$V_{i,min} = 30 + IR = 33$ V

15. $i = \dfrac{\varepsilon}{R + ki}$



$ki^2 + iR = \varepsilon$
$i^2R = \varepsilon i - ki^3$
$p = \varepsilon i - ki^3$

When p is maximum $\dfrac{dp}{di} = 0$

$p = \dfrac{dp}{di} = 0$

$\dfrac{dp}{di} = \varepsilon - 3ki^2 = 0 \quad \Rightarrow i = \sqrt{\dfrac{\varepsilon}{3k}}$

$P_{max} = \dfrac{2\varepsilon}{3}\sqrt{\dfrac{\varepsilon}{3k}}$

16.



Conserving momentum

loss in kinetic energy = $k - \dfrac{k}{4} - \dfrac{9k}{16} = \dfrac{3k}{16}$

$\dfrac{3k}{16} = 12.1 \times 4$ ($\because$ He^+ must reach its 2nd excited state)

$\dfrac{3k}{16} = 12.1 \times 4$ ($\because$ He^+, 2nd)

$k \approx 258\,eV$

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17. $\Delta L = \dfrac{FL}{AY}$

$$\Delta L_{AB} = \frac{2\times10^4\times2}{50\times10^{-4}\times4\times10^9} = \frac{1}{500}\,m$$

$$\Delta L_{BC} = \frac{4\times10^4\times2}{50\times10^{-4}\times4\times10^9} = \frac{2}{500}\,m$$

$$\Delta L_{CD} = \frac{8\times10^4\times1}{50\times10^{-4}\times4\times10^9} = \frac{2}{500}\,m$$

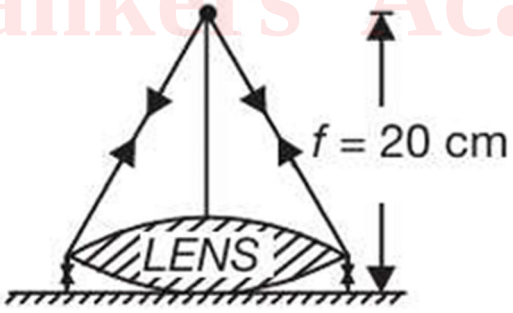
$$\Delta L = \Delta L_{AB} + \Delta L_{BC} + \Delta L_{CD}$$

$$= \frac{5}{500} = 10^{-2}\,m = 1cm$$

18.
- A Correct - Photodiodes are used in reverse bias for better sensitivity,
- B Correct – Solar cells have thick/wide depletion region to absorb maximum photons.
- C Wrong- LEDs use moderately doped or lightly doped or lightly doped junctions (heavily doped ones reduce efficiency due to non-radiative recombination).
- D Wrong – At very high intensity, photocurrent in photodiode tends to saturate.
- E Correct – Solar cells generate power without external bias (zero bias condition).

19. Focal length of lens is $f_l = 20\ cm$

Focal length of combination is $f_c = 20+5 = 25\ cm$



$$\frac{1}{f} = \frac{1}{f_l} + \frac{1}{f}$$

$$\frac{1}{25} = \frac{1}{f_l} + \frac{1}{20}$$

$$\Rightarrow f_l = -100\,cm$$

For this liquid lens

$$\frac{1}{f_l} = (\mu-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$-\frac{1}{100} = (\mu-1)\left(\frac{1}{-33} - \frac{1}{\infty}\right)$$

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$\Rightarrow (\mu - 1) = 0.33$

$\Rightarrow \mu = 1.33 = \frac{4}{3}$

20. $\frac{E_1}{E_2} = \frac{C_p}{C_v} = \gamma$

For a monoatomic gas:

$\gamma = \frac{5}{3}$

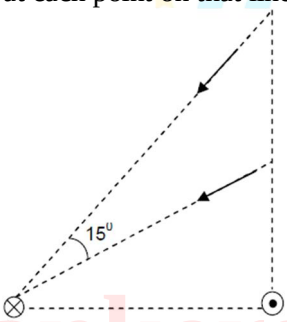
Given,

$\frac{E_1}{E_2} = \frac{5}{3} = \frac{x}{6}$

Cross-multiply:

$5 \times 6 = 3x$

21. at each point on that line magnetic fires is towards wire carrying current inwards.



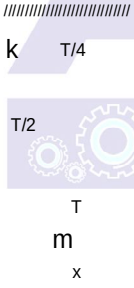
22. $U = \frac{-CGm^2}{a}$

then total G.P.E of bigger cube is U' = 32 U

So to bring work done by external agent is 24 U

23. Apply $B = \frac{\mu_0 NiR^2}{2\left(R^2 + x^2\right)^{3/2}}$

24. Let mass 'm' falls down by x so spring extends by 4x; which causes an extra tension T in lowest string



Thus the magnitude of T can be found as follows:

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$$\Rightarrow \quad \frac{T}{4} = k(4x)$$

$T = (16k)x$

Thus equation of motion of mass m is

$T = ma$

$$\Rightarrow \quad a = -\frac{16k}{m}x$$

Comparing with $a = -\omega^2 x$

we get $\omega = \sqrt{\frac{16k}{m}}$

$$\Rightarrow \quad f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{16k}{m}}$$

$$= \frac{2}{\pi}\sqrt{\frac{k}{m}}$$

25. Efficiency $\eta = \frac{W}{Q_1} = \frac{T_1 - T_2}{T_1}$

Coefficient of performance of refrigerator

$$K = \frac{Q_4}{W} = \frac{T_4}{T_3 - T_4}$$

Now vc $Q_4 = Q_3 - W$

$$\frac{Q_3 - W}{W} = \frac{T_4}{T_3 - T_4}$$

$$\frac{T_4}{T_3 - T_4} = \frac{Q_3}{W} - 1 = \frac{Q_3 T_1}{Q_1(T_1 - T_2)} - 1$$

Solving : $\frac{Q_3}{Q_1} = \left(\frac{T_4}{T_3 - T_4} + 1\right)\left(\frac{T_1 - T_2}{T_1}\right)$

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CHEMISTRY

26. Stronger acid has lower pKa value.

27. $t_{1/2} \propto \frac{1}{a^{n-1}}$

28. $2\pi r_2 = n\lambda = 2\lambda \Rightarrow n = 2$

$$E_n = -13.6 \times \frac{(2)^2}{(2)^2} = -13.6 eV / atom$$

29. $\rightarrow$ Be, Mg & N has +ve but P has –ve ΔHeg

30. A. $C_6H_5NHCOC_6H_5$ C. $C_6H_{12}O_6$

B. $CH_2 - COO^{\ominus}$ (with $NH_3^{\oplus}$) D. $(CH_3)_4 N^{\oplus}Cl^{\ominus}$

31. A, B, C give iodoform test.

32. (A) Cold, dilute alkaline $KMnO_4$

(B) $CuSO_4$ (alkaline) + complexing reagent

(C) Anhydrous $ZnCl_2 +$ Conc HCl

(D) $AgNO_3 + NH_4OH$

33. It can show linkage and geometrical isomerism.

34. Both turn lime water milky.

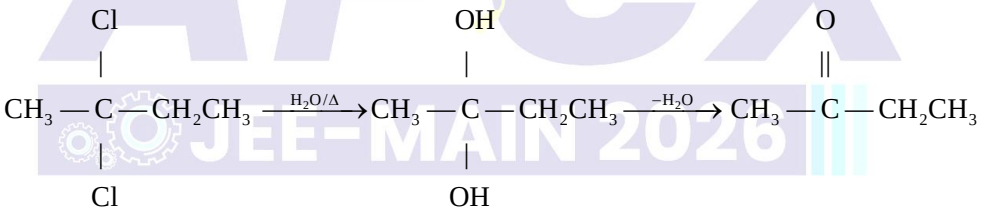
35. Products are $CO_2 + H_2O +$ Acetone

36. Both XeO_3 & SO_3^{2-} are pyramidal (non-planar).

37. Ethers are explosive due to formation of peroxide

38. $Ni(dmg)_2 \downarrow$ (Rosy red ppt)

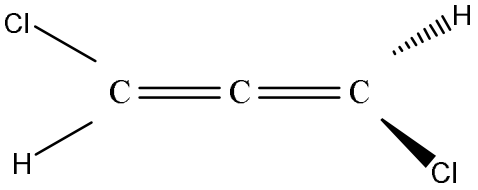
39.



40. $C_8H_6 + 9.5O_2 \longrightarrow 8CO_2 + 3H_2O$

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41.



(Chiral Without chiral centre)

42. Correct chemical formula of tear gas is CCl_3NO_2

43. For isothermal process, $\Delta T = 0 \Rightarrow \Delta U = 0, \Delta H = 0, \Delta(PV) = 0$

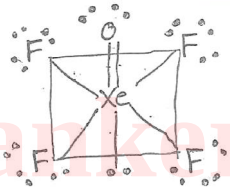
44. $K_2Cr_2O_7(s) \xrightarrow{\Delta} \underset{\text{Yellow salt}}{K_2\overset{+6}{Cr}O_4(s)} + \overset{+3}{Cr_2}O_3(s) + \underset{\text{Paramagnetic gas}}{O_2(g)}$

45. As, dissociation of H_2O is endothermic, $K_\omega \uparrow$ as $T \uparrow$

46. $pOH = pK_b + \log\frac{100\times0.1\times2}{200\times0.05} = 4.7 + \log 2 = 5$

$pH = 14 - 5 = 9$

47. $6 + 3 \times 4 + 2 + 1 = 21$



48. $P_A = 0.4 \times 50 = 20$, $P_B = 0.6 \times 100 = 60$

$\therefore x'_B = \frac{P_B}{P_T} = \frac{60}{20+60} = 0.75$

$\Rightarrow$ Mole percent of B = 75 in vapour.

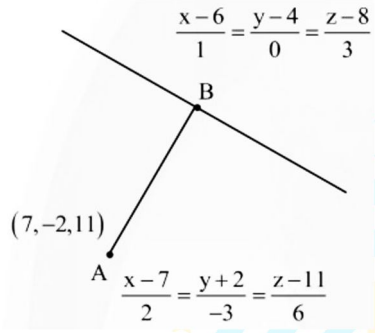
50. $E^0_{cell} = 0.83 - 0.65 = \frac{0.06}{2}\log 10^x = 0.03x$

$\therefore x = 6$

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MATHEMATICS

51. $B = (2\lambda+7, -3\lambda-2, 6\lambda+11)$



Point B lies on $\frac{x-6}{1}=\frac{y-4}{0}=\frac{z-8}{3}$

$\frac{2\lambda+7-6}{1}=\frac{-3\lambda-2-4}{0}=\frac{6\lambda+11-8}{3}$

$-3\lambda-6=0,\ \lambda=-2$

$B \Rightarrow (3,4,-1)$

$AB=\sqrt{(7-3)^2+(4+2)^2+(11+1)^2}=\sqrt{196}=14$

52. $I=\int_0^3\left|x^2-3x+2\right|dx=\int_0^3\left|\left(x-\frac{3}{2}\right)^2-\frac{1}{4}\right|dx,\ \text{let } x-\frac{3}{2}=t\Rightarrow dx=dt$

$I=2\int_0^{3/2}\left|t^2-\frac{1}{4}\right|dt=2\left[-\int_0^{1/2}\left(t^2-\frac{1}{4}\right)dt+\int_{1/2}^{3/2}\left(t^2-\frac{1}{4}\right)dt\right]=\frac{11}{6}$

53. Given R = {(a, b), {b, c}, (b, d)}
In order to make it equivalence relation as per given set, R must be {(a, a), (b, b), (c, c), (d, d), (a, b), (b, a), (b, c), (c, b), (b, d), (d, b), (a, c), (a, d), (c, d), (d, c), (c, a), (d, a)}
There already given so 13 more to be added.

54. $\left(x-\frac{1}{5}\right)^2+\left(y-\frac{2}{5}\right)^2=\left(\lambda^2-4\lambda+4\right)\left(\frac{3x+4y-1}{5}\right)^2$

If $0<|\lambda-2|<1$ i.e. $\lambda\in(1,2)\cup(3)$

55. $\cot^{-1}x=t,\ \frac{1}{1+x^2}dx=dt$

$I=\int e^t\left(\cot^2 t-\cot t+1\right)dt$

$I=\int e^t\left(\text{cosec}^2 t+(-\cot t)dt\right.$

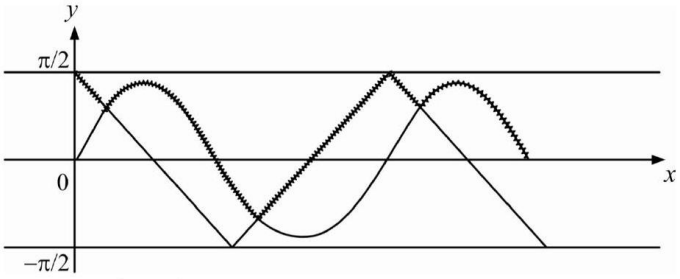
$I=-e^t\cot t+c$

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$I = -e^{\cot x} \cdot x + c$

56. $f(x) = \max\left\{\sin x, \sin^{-1}(\cos x)\right\}$

Clearly, f(x) is continuous but non-differentiable at infinite points.



57. $L = \lim\limits_{x\to 0^+}\left(\dfrac{\cot^{-1}(\ln(x))}{\pi}\right)^{-\ln x}\ (1^\infty) = e^{\lim\limits_{x\to 0^+}\left(\frac{\cot^{-1}\ln(x)-\pi}{\pi}\right)(-\ln x)} = e^{\lim\limits_{x\to 0^+}\frac{\cot^{-1}(-\ln x)}{\pi}(\ln x)}$

$= e^{-\lim\limits_{x\to 0^+}\frac{\tan^{-1}\left(-\frac{1}{\ln x}\right)}{\left(-\frac{1}{\ln x}\right)\pi}} = e^{\frac{-1}{\pi}}\ \therefore \text{L.H.L.} = f\left(0^-\right) = e^{\left(-\frac{x}{\pi},\frac{1}{x}\right)} = e^{-\frac{1}{\pi}} \Rightarrow f\left(0^+\right) = f\left(0^-\right) = e^{-\frac{1}{\pi}} = k$

58. $\int_0^1 \left(1-x^7\right)^{1/4} dx - \int_0^1 \left(1-x^4\right)^{1/7} dx$

$y = f(x) = \left(1-x^7\right)^{1/4} \Rightarrow y^4 = \left(1-x^7\right) \Rightarrow x = \left(1-y^4\right)^{1/7}$

Hence functions are inverse of each other.

$I = \int_0^1 f(x)\,dx - \int_0^1 g(y)\,dy = \int_0^1 f(x)\,dx - \int_1^0 x f'(x)\,dx$

Hence, $I = \int_0^1 \left(f(x) + x f'(x)\right) dx = x f(x)\Big]_0^1 = f(1) = 0$

59. I.F. $= e^{\int \frac{g'(x)}{g(x)}dx} = e^{\ln g(x)} = g(x)$

$$yg(x) = \frac{1}{2}\int \frac{2g(x)g'(x)}{1+g^2(x)}dx = \frac{1}{2}\ln\left[1+g^2(x)\right] + C$$

Here, $C = 1 - \dfrac{\ln 2}{2}$

$$y(e)g(e) = \frac{1}{2}\ln(1+2e-1) + 1 - \frac{\ln 2}{2} = \frac{3}{2}$$

60. Distance of z from (3 + 2i) is same as from imaginary axis $\Rightarrow$ z moves on a parabola with focus (3 + 2i) and directrix as imaginary axis.

Hence z_1 and z_2 are extremities of focal chord hence $\arg\left(\dfrac{z_1-(3+2i)}{z_2-(3+2i)}\right) = \pm\pi$

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61. $S=\frac{1\cdot3}{2}+\frac{3\cdot5}{2^2}+\frac{5\cdot7}{2^3}+\frac{7\cdot9}{2^4}+\cdots$ (1)

$\frac{1}{2}S=\frac{1.3}{2^2}+\frac{3.5}{2^3}+\frac{5.7}{2^4}+\cdots$ (2)

Subtract equation (1) and (2)

$\frac{S}{2}=\frac{3}{2}+3+\frac{5}{2}+\frac{7}{2^2}+\ldots(3)$

Let $S'=3+\frac{5}{2}+\frac{7}{2^2}+\cdots(4)$

$\frac{S'}{2}=\frac{3}{2}+\frac{5}{2^2}+\cdots$ (5)

subtracting eq(4) – eq(5)

$\frac{S'}{2}=3+1+\frac{1}{2}+\frac{1}{2^2}+\cdots$

$S'/2=3+\frac{1}{1-1/2}=3+2=5$

$\Rightarrow S'=10,\ \text{so}\ \frac{S}{2}=\frac{3}{2}+10=\frac{23}{2}\Rightarrow s=23$

62. $\left(\frac{1}{1}+\frac{1}{2}\ldots\ldots\frac{1}{17}\right)+\left(\frac{1}{1\times2}+\frac{1}{1\times3}\right)+\ldots\ldots\left(\frac{1}{1\times2\times3\ldots17}\right)$

$=\frac{(1+2+3\ldots\ldots..17)+(1\times2+1\times3+\ldots)\ldots\ldots+(1\times2\ldots\ldots16)+1}{1\times2\times3\ldots17}$

$=\frac{\sum1+\sum1.2+\sum1.2.3+\ldots\ldots+\sum1.2\ldots\ldots16+1}{1\times2\times3\ldots17}$

$=\frac{(1+1)(1+2)(1+3)\ldots\ldots(1+17)-(1\times2\ldots17)}{1\times2\times3\ldots17}$

$=\frac{1\cdot2\cdot3\ldots\ldots\ldots18-1\cdot2.3\ldots\ldots17}{1\times2\times3\ldots17}=\frac{17!(18-1)}{17!}=17$

63. $T_n=\frac{\left(r^2+r-1\right)\cdot r!}{\left(r^2+3r+2\right)}=\frac{\left[(r+1)^2-(r+2)\right]\cdot r!}{\left(r^2+3r+2\right)}=\left[\frac{(r+1)\cdot(r+1)!-r!(r+2)}{(r+1)(r+2)}\right]$

$T_r=\frac{(r+1)!}{r+2}-\frac{r!}{(r+1)}\Rightarrow T_r=V_r-V_{r-1}$

$S_n=V_n-V_0$

$S_n=\frac{(n+1)!}{n+2}-\frac{1}{2}\Rightarrow S_n=\frac{2(n+1)!-(n+2)}{2(n+2)}$

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64. $\sum_{k=0}^{4}\left(\frac{5^{4-k}}{(4-k)!}\right)\left(\frac{x^k}{k!}\right)=\sum_{k=0}^{4}\left(\frac{5^{4-k}}{(4-k)!}\right)\left(\frac{X^k}{k!}\right)\frac{4!}{4!}$

$=\sum_{k=0}^{4}\frac{{}^4C_k\cdot 5^{4-k}\cdot x^k}{4!}=\frac{(5+x)^4}{4!}$

Now $\frac{(5+x)^4}{4!}=\frac{8}{3}\Rightarrow(5+x)^4=64$

$x+5=2^{3/2}=2\sqrt{2}\Rightarrow x=2\sqrt{2}-5$

65. We have, $\frac{dy}{dx}+x^3y=x^7\ \therefore$ I.F $=e^{x^4/4}$

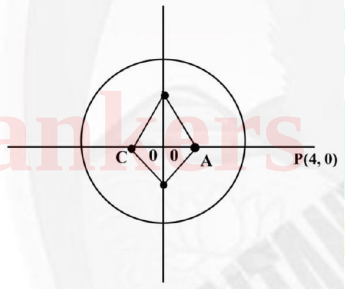
So, solution is $ye^{x^4/4}=x^4e^{x^4/4}-4e^{x^4/4}+c$

$y=x^4-4+ce^{-x^4/4}$

When y = 4, x = 0, we have $4=-4+c\Rightarrow c=8$

Now, $\left[\frac{dy}{dx}\right]_{x=\sqrt{2}}=8\sqrt{2}\left(1-\frac{2}{e}\right)$

66.



A = (2, 0), P = (4, 0)

S·D = AP = 2

67. $\because\left(1+2x^2-x^3\right)^8$

$=\sum\frac{8!}{\alpha!\beta!\gamma!}\cdot(2)^{\beta}\cdot(-1)^{\gamma}\cdot x^{2\beta+3\gamma}$

$\Rightarrow 2\beta+3\gamma=12$ and $\alpha+\beta+\gamma=8$

α	β	γ
2	6	0
3	3	2
4	0	4

$\therefore$ Coefficient of x^{12} is

$\frac{8!}{2!6!0!}(2)^6+\frac{8!}{3!3!2!}(2)^3\cdot(-1)^2+\frac{8!}{4!0!4!}(2)^0(-1)^4$

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$= 1792 + 4480 + 70 = 6342$

68. $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\cot^{-1}\left(r^2+\frac{3}{4}\right)=\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\cot^{-1}\left[\frac{(r-1/2)(r+1/2)+1}{(r+1/2)-(r-1/2)}\right]$

$=\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\cot^{-1}\left(r-\frac{1}{2}\right)-\cot^{-1}\left(r+\frac{1}{2}\right)$

$\left[\because \cot^{-1}\left(\frac{ab+1}{b-a}\right)=\cot^{-1}a-\cot^{-1}b\right]$

$=\lim\limits_{n\to\infty}\left[\cot^{-1}\left(\frac{1}{2}\right)-\cot^{-1}\left(\frac{3}{2}\right)+\cot^{-1}\left(\frac{3}{2}\right)-\cot^{-1}\left(\frac{5}{2}\right)+\ldots.+\cot^{-1}\left(n-\frac{1}{n}\right)-\cot^{-1}\left(n+\frac{1}{2}\right)\right]$

$=\lim\limits_{n\to\infty}\left[\cot^{-1}\left(\frac{1}{2}\right)-\cot^{-1}\left(n+\frac{1}{2}\right)\right]$

$=\cot^{-1}\left(\frac{1}{2}\right)-0=\tan^{-1}2$

69. $(22)^{2022}+(2022)^{22}$

Divided by 3,

$(21+1)^{2022}+(2022)^{22}$

$=3k+1(\text{ Remainder}=1)$

Divided by 7

$(21+1)^{2022}+(2023-1)^{22}\quad[\because As2023=7\times289]$

$7k+2(\beta=2)$

So, $\alpha^2+\beta^2=5$

70. Differentiate given function w.r.t. x both sides.

$\Rightarrow f'(x)=\frac{2x}{x^2+1}+e^{-x}>0,\forall x\in R$

For every real value f(x) is strictly increasing.

Given function $g(x)=\frac{1-2e^{2x}}{e^x}=e^{-x}-2e^x$

Differentiate w.r.t. x both sides,

$\Rightarrow g'(x)=-\left(2e^x+e^{-x}\right)<0,\forall x\in R\Rightarrow g$ is decreasing.

Take $f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right)>f\left(g\left(\alpha-\frac{5}{3}\right)\right)$

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$\Rightarrow g\left(\frac{(\alpha-1)^2}{3}\right) > g\left(\alpha-\frac{5}{3}\right) \Rightarrow \frac{(\alpha-1)^2}{3} < \alpha-\frac{5}{3}$

$\Rightarrow \alpha^2-5\alpha+6<0 \Rightarrow (\alpha-2)(\alpha-3)<0 \Rightarrow \alpha \in (2,3)$

71. We know that, if $\vec{r_1}=\vec{a}+\lambda\vec{b}$ and

$\vec{r_2}=\vec{c}+\mu\vec{d}$, then shortest distance between two lines is given by $\frac{|(\vec{b}\times\vec{d})\cdot(\vec{a}-\vec{c})|}{|\vec{b}\times\vec{d}|}$

Now, $\vec{a}-\vec{c}=(\alpha+4)\hat{i}+2\hat{j}+3\hat{k}$

Also, $\vec{b}\times\vec{d}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -2 & 2\\ 3 & -2 & -2\end{vmatrix}=8\hat{i}+8\hat{j}+4\hat{k}$

$\therefore |\vec{b}\times\vec{d}|=12$

Now, $\frac{\vec{b}\times\vec{d}}{|\vec{b}\times\vec{d}|}=\frac{2\hat{i}+2\hat{j}+\hat{k}}{3}$

$\therefore$ Required distance

$=\frac{\left((\alpha+4)\hat{i}+2\hat{j}+3\hat{k}\right)\cdot\left(2\hat{i}+2\hat{j}+\hat{k}\right)}{3}=9$

$=2\alpha+8+4+3=27 \Rightarrow 2\alpha=27-15$

$=2\alpha=12 \Rightarrow \alpha=6$

72. $1+x+x^2=0 \Rightarrow \alpha=\omega$ or ω^2 if $\alpha=\omega$

So, $(1+\omega)^7=a+b\omega+c\omega^2 \Rightarrow -\omega^2=a+b\omega+c\omega^2$

$\Rightarrow a=0,b=0,c=-1 \therefore 5(3a-2b-c)=5$

73. x, x, x, y, y

$AM \geq GM$

$\frac{3x+2y}{5} \geq \left(x^3y^2\right)^{1/5}$

$\frac{3x+2y}{5} \geq \left(2^{15}\right)^{1/5}$

$3x+2y \geq 5\cdot 2^3$

$3x+2y \geq 40$

74. using expansions

$\alpha-\beta=0, \alpha+\frac{\beta}{2}+\gamma=0, \frac{\alpha}{2}-\frac{\beta}{2}-\gamma=10$

Solving $\alpha=6,\beta=6,\gamma=-9$

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$\alpha+\beta+\gamma=3$

75. $A=\begin{bmatrix}2 & -1 & 1\\ -1 & 2 & -1\\ 1 & -1 & 2\end{bmatrix}, \det A=4$

$\left|\left(\text{adj}5\left(2\left(A^{-1}\right)\right)\right)\right|=\left|3.2^2\,\text{adj}\left(A^{-1}\right)\right|$

$=12^3\left|\text{adj}A^{-1}\right|=12^3\left|A^{-1}\right|$

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